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DEPARTEMENT	FINANCE ACCOUNTING LAW
PROGRAM	MSc CFIB
MODULE NAME	ENTERPRISE RISK MANAGEMENT
MODULE Leader	Henri TRAN
NAME(s) of grading professor(s)	Foly ANANOU

FINAL EXAMEN

Exam date: April 15th

Duration: 2h

NOM, First name of the author	Foly ANANOU
REVIEWED by Henri TRAN	Date: visa:

INSTRUCTIONS:

- Only simple handheld calculators are allowed. Phones, computers, and programmable calculators are **NOT** permitted.
- No course teaching materials are allowed (closed book).
- Show your steps for all computations. Unsupported numerical answers receive no credit.
- Clarity of reasoning matters more than length — be concise and rigorous.

Structure:

- Part I — Risk concepts & ERM governance: **25 points** (≈30 min)
- Part II — Measurement & hedging (computations): **40 points** (≈55 min)
- Part III — ERM in practice & cases: **35 points** (≈35 min)

Part I — Risk Concepts & ERM Governance

(25 points)

Question 1

(12 points)

(a) Risk vs. uncertainty (6 pts). Explain the difference between *risk* and *uncertainty* in the ERM context. Your answer must address: (i) what can and cannot be reliably quantified, and (ii) why scenario analysis complements (rather than replaces) quantitative risk metrics.

(b) Why silo risk management fails (6 pts). Explain why managing risks in silos can increase enterprise-wide vulnerability, even when each risk category appears well controlled locally. Support your argument with a concrete example involving at least two risk categories (e.g. market + liquidity; credit + funding; operational + reputational).

Question 2

(13 points)

Consider the statement:

“ERM fails when risk information exists but does not change decisions.”

Propose an ERM architecture that increases the probability that risk signals will affect decisions. Your answer must address:

- the distinction between risk *ownership* and risk *oversight*,
- escalation thresholds and exception governance,
- why independence of the risk function is necessary but not sufficient.

Part II — Measurement & Hedging

(40 points)

Question 3 — RAROC and EVA

(10 points)

A business unit proposes a new portfolio with the following annual figures (in € million):

$$NI = 28 \quad EC = 200 \quad k = 11\%$$

where NI is net income after expected losses, EC is economic capital allocated at the chosen confidence level, and k is the firm-wide hurdle rate. For EVA purposes, assume invested capital equals economic capital.

- (a) Define **RAROC** and compute it for this portfolio. (3 pts)
- (b) Define **EVA** and compute it. (3 pts)
- (c) A manager concludes: “ $RAROC > k \Rightarrow$ always accept.” Give two reasons why this rule can be misleading in ERM practice. (4 pts)

Question 4 — Hedging: prospective vs. retrospective effectiveness + tail risk

(30 points)

A firm hedges a monthly exposure using a derivative. Let X be the unhedged monthly P&L (€m) and H the monthly P&L of the hedging instrument. The hedged position is $Y = X - hH$. The hedge ratio h is set using normal-period estimates and kept fixed during stress.

	Period	$\text{Var}(X)$	$\text{Var}(H)$	$\text{Cov}(X, H)$
Data (monthly, €m)	Normal (estimation)	100	64	48
	Stress (realized)	225	144	12

Hedge effectiveness is defined as:

$$HE = 1 - \frac{\text{Var}(Y)}{\text{Var}(X)}, \quad \text{Var}(Y) = \text{Var}(X) + h^2 \text{Var}(H) - 2h \text{Cov}(X, H)$$

- (a) Compute the minimum-variance hedge ratio: $h^* = \frac{\text{Cov}(X, H)}{\text{Var}(H)}$ using normal-period data. (4 pts)
- (b) Using $h = h^*$, compute $\text{Var}(Y)$ and HE_{pros} in the normal period. (5 pts)
- (c) Keeping $h = h^*$, compute $\text{Var}(Y)$ and HE_{retro} using stress-period data. (5 pts)
- (d) Interpret the change from HE_{pros} to HE_{retro} . Your answer must reference *regime change* and at least one of: basis/correlation breakdown, model risk, or liquidity/market functioning. (4 pts)
- (e) **Tail-risk check.** In stress months, the unhedged loss distribution of X is:

$$L_X \in \{0, 5, 10, 25, 40\} \quad \text{with probabilities} \quad \{0.40, 0.25, 0.20, 0.10, 0.05\}$$

(Positive values are losses, in €m.)

- (i) Compute $\text{VaR}_{95\%}(L_X)$. (3 pts)
- (ii) Compute $\text{ES}_{95\%}(L_X)$ (expected loss conditional on exceeding $\text{VaR}_{95\%}$). (5 pts)
- (iii) Explain in one or two sentences why ES is more informative than VaR for tail-risk management. (4 pts)

Part III — ERM in Practice & Cases

(35 points)

Question 5 — Cross-industry ERM

(15 points)

Compare ERM in two industries (choose two: financial institutions, energy & commodities, non-financial corporations, public sector & infrastructure). For each industry, explain in no more than one short paragraph:

- how industry economics shape ERM priorities,
- the main instruments used (models/capital, hedging, operational controls, etc.),
- a typical failure mode specific to that industry.

Conclude with one sentence on the key difference between the two industries you selected.

Question 6 — Case: “ERM existed, crisis still happened”

(20 points)

A large firm has adopted a recognized ERM framework, produces regular risk reports, and uses quantitative risk metrics. Despite this, it experiences a major crisis driven by a combination of market stress, liquidity pressure, and delayed escalation.

Diagnose the failure. Your answer must:

- distinguish *measurement failure* from *governance failure* (with one example of each),
- explain the role of incentives in delayed escalation,
- explain why “more data” or “better models” would not necessarily have prevented the crisis,
- propose two concrete architecture or process changes that would reduce recurrence.